

A Comparative Study on Machine Learning and Deep Learning Approaches for Cybersecurity Intrusion Detection System

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Abstract— This paper illustrates a study of various machine learning and deep learning state-of-the-art approaches for a cyber security intrusion detection system. As datasets pertaining to the field of Cybersecurity are very tough to get one's hands on as they are extremely rare or are not publicly available due to the concerns for the privacy of the data, the CSE-CIC-IDS 2018 dataset has been generated as a part of a collaborative project between the Communications Security Establishment (CSE) and the Canadian Institute for Cybersecurity (CIC), which is the dataset used for this paper. This paper provides a detailed study on the comparisons for the different machine learning and deep learning approaches used to build an intrusion detection system using this dataset.

Keywords—component machine learning, deep learning, intrusion detection system, CSE-CIC-IDS 2018, feedforward neural network, logistic regression, decision tree classifier, support vector classifier

I. INTRODUCTION

In this time of digital age, technology has covered almost every aspect of our life. It has penetrated almost all fields of study. One can say that our life now revolves around it entirely. From electric toothbrushes, digital books to holographic concerts like at the ABBA Arena, London, giving almost next to real experiences. Simultaneously, the use of laptops, desktop PCs, smartphones, tablets, etc. has reached to an extreme peak from our personal life and professional life. As the role of technology in our life in this globalised digital age has risen, so has risen the risks of using it. There have been numerous cyber hacks in the past couple of years leading to database leaks, ransom situations and a lot more. For instance, Dharma ransomware using RDP intrusion¹, Zoho ManageEngine zero-day attack² etc. Some other types of attacks on networks are Denial-of-Service, Botnet attacks, SQL Injection, Bruteforce attacks, Distributed Denial-of-Service (DDOs) and other cyberattacks. While using technology has eased people's lives in plenty of ways, it has endangered their privacy as well. Data privacy is becoming a huge concern in light of the recent cybersecurity hacks. The data leaks around the world have left a lot of people exposed people's private and confidential data because of which so many people are likely targets to huge number of scams through various phishing techniques, may it be conducted via

emails, voice calls or else. These attacks don't just make use of people's data or scam people, but also a few attacks have its counter effects on people's lives too where deaths have been a result indirectly due to a hack during a cyberattack. According to an article³ published by Patrick Howell O'Neill, as a result of a ransomware attack by hackers on a German Hospital, a female patient had suffered death indirectly. As the systems in the hospital were under a ransomware attack, she had to be transferred to a different hospital as the hospital under attack was not in a capability to provide her enough care for her treatment and survival, and due to this, she passed away. In cybersecurity, most of the tasks can be completed at a much faster pace with use of automation and machine learning compared to a completely manual approach. Automation can be used for repetitive tasks in order to reduce the human force for that task. On the other hand, machine learning can be used for constantly learning through analysis of data for finding patterns and anomalies in order to detect malware in encrypted traffic, for finding insider threats, and also warning people to be safe when using internet in a dangerous locality, and even guard safely stored data on cloud by detecting sceptical changes in a user behaviour. Thus, if the analysis is automated, cybersecurity agents can quickly uncover threats and seclude events that need deeper human analysis. Below is a list of state-of-the-art use cases for machine learning in cybersecurity by CISCO:

How machine learning helps security...	
Find threats on a network	Machine learning detects threats by constantly monitoring the behavior of the network for anomalies. Machine learning engines process massive amounts of data in near real time to discover critical incidents. These techniques allow for the detection of insider threats, unknown malware, and policy violations.
Keep people safe when browsing	Machine learning can predict "bad neighborhoods" online to help prevent people from connecting to malicious websites. Machine learning analyzes Internet activity to automatically identify attack infrastructures staged for current and emergent threats.
Provide endpoint malware protection	Algorithms can detect never-before-seen malware that is trying to run on endpoints. It identifies new malicious files and activity based on the attributes and behaviors of known malware.
Protect data in the cloud	Machine learning can protect productivity by analyzing suspicious cloud app login activity, detecting location-based anomalies, and conducting IP reputation analysis to identify threats and risks in cloud apps and platforms.
Detect malware in encrypted traffic	Machine learning can detect malware in encrypted traffic by analyzing encrypted traffic data elements in common network telemetry. Rather than decrypting, machine learning algorithms pinpoint malicious patterns to find threats hidden with encryption.

Figure 1. Role of Machine learning in Security⁴

¹ Available from <https://darktrace.com/blog/old-but-still-dangerous-dharma-ransomware-via-rdp-intrusion> [online]

² Available from <https://threatpost.com/zoho-zero-day-manageengine-active-attack/177178/> [online]

³ Available from <https://www.technologyreview.com/2020/09/18/1008582/a-patient-has-died-after-ransomware-hackers-hit-a-german-hospital> [online]

⁴ Available from <https://www.cisco.com/c/en/us/products/security/machine-learning-security.html> [online]

Cyber security was never an aim for the traditional Machine Learning techniques as also mentioned by K. Shaukat et al. (2020)⁵, therefore, there are various challenges for the use of Machine Learning in Cybersecurity. It can be noted that machine learning can also be used Cybersecurity for several use cases in like malware analysis, vulnerability discovery, botnet detection and containment, insider threat, binary code analysis (which is useful when there is no direct access to the code), threat assessment, vulnerability testing, intrusion detection system and so on. Nonetheless, Machine Learning methods need a significant number of high-performance resources and an ample amount of data for the model training purposes. Thus, there is a requirement to possess highly capable and vigorous Machine Learning techniques which are specially designed for managing cybersecurity attacks, that too which can detect those real-time threats in short intervals. However, avoiding cybersecurity attacks at a very early stage is one more tricky provocation. Additionally, early and rapid detection of new and zero-day (or 0-day) attacks is even now a very demanding niche for research, where a zero-day attack means an attack using a software vulnerability discovered by hackers before the vendor has become aware of it; as the vendor is unaware about the vulnerability, they had 'zero-days' to fix it, and hence the name – 'zero-day' attack.

Also, in order to cover all the grounds of organizational, legal and technical grounds for cybersecurity, a lot of specific security measures are also required along with regulations, directives and policies. Intrusion detection systems act as an important part of the defence line of a system. However, one needs other security precautions along with an Intrusion Detection System like authorized access control, authentication systems and cryptographic methods to have an even more secured system against cyberattacks.⁶

But what is an Intrusion Detection System? As explained by Check Point Research, an intrusion detection system (IDS; also, intrusion prevention system or IPS) is a device or software application that keeps a track of a network or systems for suspicious, harmful or malicious activity or policy infringements⁷. Furthermore, the different types of Intrusion Detection System vary from single/personal computers to large networks. Furthermore, the two most common classification for an IDS are: a Host-based Intrusion Detection System and a network Intrusion Detection System.

- **Host-Based Intrusion Detection System (HIDS):** A host-based IDS is stationed on a specific endpoint and prepared with the purpose of securing it in opposition to external and internal threats. An IDS's range of vision is restricted to the device of the host, which further reduces circumstances for decision-making when it is created mainly focusing on the hosts, but has deep perceptibility into the host computer's internals. This type of an IDS might have the potential to keep track of the flow of networks flowing towards and away from the machines, check ongoing operations, and examine the records of the system⁸.

- **Network-Based Intrusion Detection System (NIDS):** A network-based IDS is devised to keep an eye on the completely secured network. Having provided the complete access for the network flow requests flowing through the entire network, decision making takes place depending on the data about the data and in the data inside the packets in the network flow. Thus, such a wider perspective gives even more factors and the skill to ascertain extensive threats. Nevertheless, these systems face the scarcity of vision inside the interiors of the endpoints which are guarded by them and so, it cannot protect the system from internal threats as mentioned by M A Ferag et al. (2020)⁵.

Furthermore, there are 2 types of NIDSs, signature-based detection and anomaly-based detection. The signature-based NIDS uses a database of known threats to detect them, however, it is not possible to detect new attacks using this type of system as it is restricted to detect only known attacks, and thus, it is blind to 0-day attacks. On the other hand, anomaly-based NIDS builds a model by studying the normal or the benign behaviour of users in the protected system. After the training of the model, all future behaviour is compared to this previous normal behaviour and all anomalies are pointed out with alerts as potential threats. As this model tracks the behaviour of the network system, it is possible to detect zero-day attacks using this method.

In this research paper, a Network-based Intrusion Detection System has been discussed with comparisons on various Machine Learning and Deep Learning models. The CSE-CIC-IDS 2018 dataset has been used for the respective training the different models in context of this research paper as IDSs can distinguish between normal and malicious actions using patterns of benign traffic or normal behaviour or specific rules that describe a specific attack.

II. BACKGROUND – LITERATURE REVIEW

In the literature, there are different related studies that deal with machine learning techniques for intrusion detection systems. It is seen that network-based intrusion detection system has been intensively researched these days. The first IDS was based on research by Dorothy E. Denning (1986)⁹. It provides a system called Intrusion Detection Expert System. It had a dual approach with an expert system based on various rules in order to detect the known types of intrusions and also a statistical anomaly detection component based on user profiles, host systems and the target systems. The authors of the paper Z. Jadidi et al. (2013)¹⁰ have proposed a network-based anomaly detection system which can classify benign or harmful malicious network traffic with an extremely good accuracy. This system is created using a Multi-Layer Perceptron and Gravitational Search Algorithm. This system can categorize normal and harmful flows of network traffic with an exceedingly good accuracy rate. P. Winter et al. (2011)¹¹ proposed a NIDS using a support vector machine with a single class classifier for its exploration. There is a unique characteristic about this work; they trained their model

⁵ K. Shaukat et al. (2020)

⁶ M A Ferag et al. (2020)

⁷ Available from <https://www.checkpoint.com/cyber-hub/network-security/what-is-an-intrusion-detection-system-ids/#DefinitionofIDS> [online]

⁸ Available from <https://www.checkpoint.com/cyber-hub/network-security/what-is-an-intrusion-detection-system-ids/> [online]

⁹ D. E. Denning (1986)

¹⁰ Z. Jadidi et al. (2013)

¹¹ P. Winter et al. (2011)

using a malicious network dataset rather than a dataset with normal behaviour, and they even achieved to get a low false alarm rate.

Usually, availability of datasets is excessively scarce due to two main reasons. Firstly, datasets pertaining to security are not easily shared or made public due to privacy concerns, which results in not enough data available for research. On the contrary, most of the data in the available datasets is made available with extreme anonymity and thus, it does not reflect some trends and deficits from certain statistical characteristics¹². However, network behaviour, patterns and intrusions change and evolve with usage and passage of time, it has been evident that it is time to choose more dynamically generated datasets over the static datasets.

III. EXPERIMENTS

A. Dataset

It is seen that majority of the research carried out under this niche uses datasets like KDD CUP 1999, NSL KD'99, Kyoto 2006+, UNSW NB15, ISCX-2012, NSL KDD, CIDD001, CICIDS2017 and many more other datasets⁵. A lot of research has already been done on these datasets and so, one of the latest datasets which has not had much work done upon it is used here in this paper, which is the CSE-CIC-IDS 2018 dataset. This dataset is a part of a project carried out as a collaboration between 2 institutes namely, Communications Security Establishment (CSE) and the Canadian Institute of Cybersecurity (CIC). The attacking infrastructure used here comprises of 50 machines whereas the target group covers a total of 5 different departments covering covers 420 machines and 30 servers. This dataset incorporates the captured flow of network requests and records of the for each machine, in addition to 80 different attributes drawn out of the recorded flow of requests using CICFlowMeter-V3¹³. This dataset was generated over 10 days recording network traffic flow for that much amount of time. Almost each day was used for different types of attacks on the system and data was collected accordingly into 10 different files, one for each day.

CICFlowMeter is a software written in java which is used as generator for network traffic flow, and it also provides more flexibility for selecting the features needed to calculate, including new features, and also having an improved control of the timeout duration for the network flow. It generates two-directional Flows (Biflow), where the initial packet chooses to move forward (source to destination) and to move backward (destination to source) directions. Thus, the 83 quantitative features such as Duration, Number of pkTs, Number of byts, Length of packets, etc. are also computed individually in the forward and also in the Backward direction, which we can see in the fourth dataset file.

The output of the use of the CICFlowMeter is stored in excel file format i.e. 'CSV' for further usage with six columns named for each flow, namely FlowID, SourceIP, DestinationIP, SourcePort, DestinationPort, and Protocol with more than 80 network traffic features. It can be noted that usually the TCP flows are discontinued upon connection

teardown (by FIN packet) whereas the UDP flows are stopped by a flow timeout.

B. Preprocessing

Upon studying the dataset thoroughly, it is seen that the dataset has a lot almost 1.5 million records in each csv file with a huge bias towards the benign attacks. In order to fairly train the respective Machine Learning and Deep Learning models, the number of records for the respective attacks and benign attacks are balanced equivalently so that the model doesn't favour any type more than the other considering the number of samples in training.



Figure 2. Word Cloud for all the different types of cyberattacks described in the dataset.

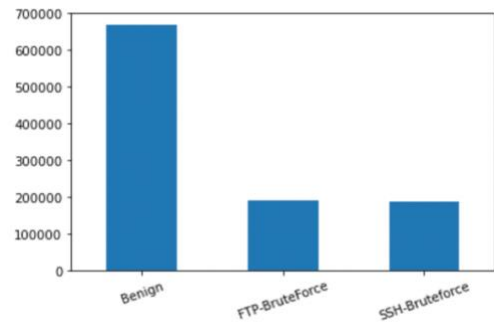


Figure 3. Data Exploration for Benign vs FTP-BruteForce and SSH-BruteForce types of cyberattacks.

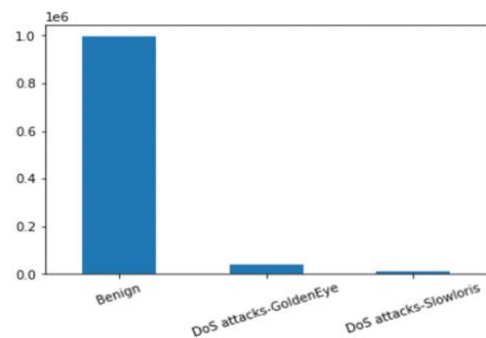


Figure 4. Data Exploration for Benign vs DoS-GoldenEye and DoS-SlowLoris types of cyberattacks.

Furthermore, all specific attacks are then generalised into their specific categories of BruteForce, DDoS, DoS, and other

¹² Available from <https://www.unb.ca/cic/datasets/ids-2018.html> [online]

¹³ Available from <https://www.unb.ca/cic/datasets/ids-2018.html> [online]

respective classes to ease the work for our model as shown in the images for the first two datasets.

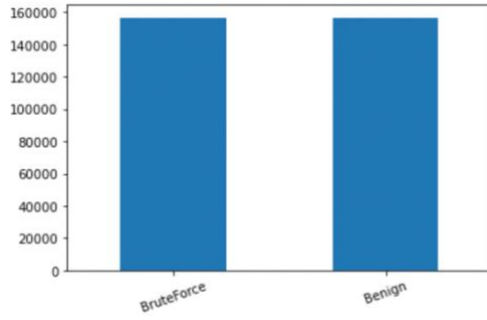


Figure 5. Benign vs BruteForce - Data Exploration after the preprocessing of the 1st dataset out of all 10 data files

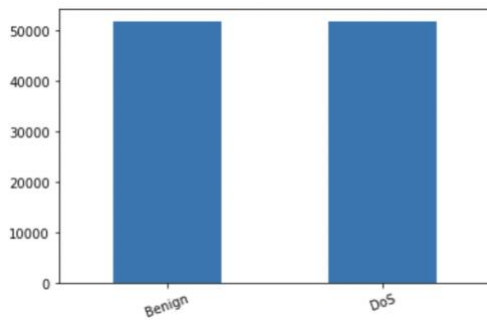


Figure 6. Benign vs DoS - Data Exploration after the preprocessing of the 2nd dataset

However, for ease of calculations, all the different types of cyberattacks that are described in the dataset are set to ‘harmful’ using the Label Encoder available from the preprocessing module of the sklearn package in Python.

Moreover, it is checked if there are any null or nan values in the dataset and all records with such values are dropped as it will hamper the training process of our models later. Furthermore, to ease the process of training our model, we encode the labels to a unique number individually.

C. Feature Selection

Considering all the 80 features for training the model would figuratively take an entire lifetime to process. Also, a lot of them would be redundant as one or the other feature might be related to another feature and thus, might not add its value to the training process. Such features can be eliminated from being used for the training process. For this purpose, correlation among all the features is calculated and can also be seen in the heatmap shown here. Furthermore, all the features with correlation equal to or more than 0.9 are dropped. This results in a list of 48 features excluding the Label column remaining in our dataset all of which are mentioned in the table below:

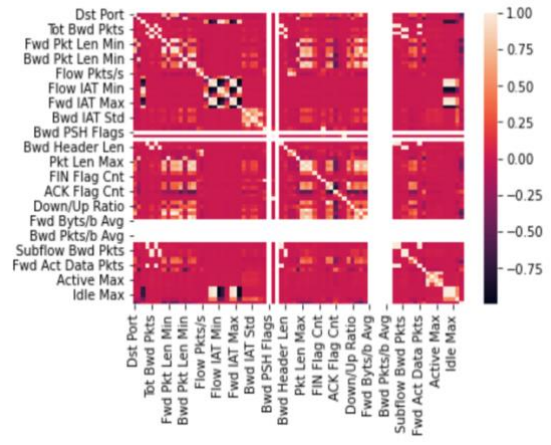


Figure 7. Heatmap for Correlation on the entire Dataframe

Feature Name	Description
Dst Port	The Port number for the Destination of the network traffic
Protocol	Protocol for the respective network traffic
Flow Duration	Duration of Flow for the network traffic
Tot Fwd Pkts	The total number of packets flowing forwards
Tot Bwd Pkts	The total number of packets flowing backwards
Fwd Pkt Len Max	The largest size of the packet flowing forwards
Fwd Pkt Len Min	The smallest size of the packet flowing forwards
Bwd Pkt Len Max	The largest size of the packet flowing backwards
Bwd Pkt Len Min	The smallest size of the packet flowing backwards
Bwd Pkt Len Mean	The mean of the sizes of all packets flowing backwards
Flow Bytes/s	The rate of flow of Bytes i.e. number of total bytes flowing in every second
Flow Pkts/s	The rate of flow of Packets i.e. number of total packets flowing in every second
Flow IAT Mean	The mean of the time taken between 2 consecutive flows
Flow IAT Std	The standard deviation of the time taken between 2 consecutive flows

Flow IAT Max	The longest duration of time between 2 consecutive flows
Flow IAT Min	The shortest duration of time between 2 consecutive flows
Bwd IAT Tot	The aggregate of the time between 2 consecutive packets flowing backwards
Bwd IAT Mean	The average of the time between 2 consecutive packets flowing backwards
Bwd IAT Std	The standard deviation of the time between 2 consecutive packets flowing backwards
Bwd IAT Max	The longest duration of time between 2 consecutive packets flowing backwards
Bwd IAT Min	The shortest duration of time between 2 consecutive packets flowing backwards
Fwd PSH Flags	The total count of the PSH flag being set inside the packets flowing forwards (where, UDP is 0)
Bwd PSH Flags	The total count of the PSH flag being set inside the packets flowing backwards (where, UDP is 0)
Fwd URG Flags	The total count of the URG flag being set inside the packets flowing forwards (where, UDP is 0)
Bwd URG Flags	The total count of the URG flag being set inside the packets flowing backwards (where, UDP is 0)
Bwd Pkts/s	The rate of flow of Packets i.e. number of total packets flowing backwards in every second
Pkt Len Max	The largest size of the packet flowing
Pkt Len Var	The minimum duration of time for packets between arrivals
FIN Flag Cnt	The total count of the FIN flag being set inside the packets flowing
RST Flag Cnt	The total count of the RST flag being set inside the packets flowing

PSH Flag Cnt	The total count of the PUSH flag being set inside the packets flowing
ACK Flag Cnt	The total count of the ACK flag being set inside the packets flowing
URG Flag Cnt	The total count of the URG flag being set inside the packets flowing
Down/Up Ratio	The ration for Download to upload for the packets
Fwd Byts/b Avg	The mean of the total count of bytes bulk rate flowing forwards
Fwd Pkts/b Avg	The mean of the total count of packets bulk rate flowing forwards
Fwd Blk Rate Avg	The mean of the total count of bulk rate flowing forwards
Bwd Byts/b Avg	The mean of the total count of bytes bulk rate flowing backwards
Bwd Pkts/b Avg	The mean of the total count of packets bulk rate flowing backwards
Bwd Blk Rate Avg	The mean of the total count of bulk rate flowing backwards
Init Fwd Win Byts	The total count of the bytes directed to flow forwards in the initial window
Init Bwd Win Byts	The total count of the bytes directed to flow backwards in the initial window
Fwd Seg Size Min	The smallest size of the segment seen flowing forwards
Active Mean	The average duration of the time for when a flow being active before it becomes idle
Active Std	The standard deviation of the time duration for when a flow being active before it becomes idle
Idle Min	The minimum duration of the time for when a flow being active before it becomes idle

The data from the dataset features is then scaled using the MinMaxScaler function from the Preprocessing module available in the scikit-learn package for python.

IV. MACHINE LEARNING AND DEEP LEARNING MODELS

Initially, the deep learning model, Feed Forward Neural Network is trained and tested after which the other models would be trained and compared to it. The Feed Forward Neural Network Model, here in use, is comprised of three layers. The inputs from the dataset are passed to the initial Dense layer which comprises of 128 nodes. The activation function used for this, and the next layer is the same function, 'ReLU' which stands for 'Rectified Linear Unit'. One of the main benefits of using a ReLU function is that all neurons are not activated at the same time when it is used. The next Dense layer which is also a hidden layer, is connected with a dropout layer with a dropout rate 0.2. The second Dense layer in concern comprises of 64 nodes which also has a similar activation function, the ReLU (Rectified Linear Unit). This layer is then connected to another Dropout layer with the same dropout rate 0.2, which then further connects it to another Dense layer. This is the third and final dense layer with 32 nodes and a similar activation function, i.e. ReLU which again is connected to a dropout layer with the same dropout rate as before. All of this is then finally connected to the outmost layer, which is the output layer. Here, the output layer comprises of 6 nodes as our dataset reads the attacks into 6 categories, namely Benign, Brakforce, Denial-of-Service (DoS), Distributed Denial-of-Service (DDoS), Botnet and other attacks like SQL Injection, Infiltration and so on. Hence, the activation function for this layer to provide a solution for this multiclassification problem is the 'softmax' function as it provides the probability for each of the classification possibility for the respective flow of traffic.¹⁴

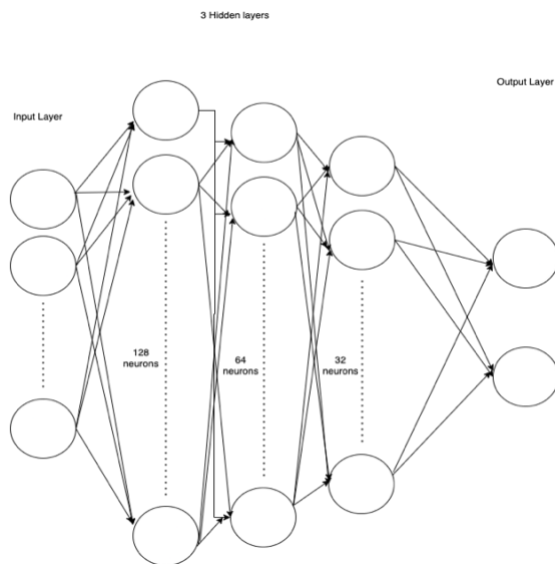


Figure 8. Feed Forward Neural Network

This dataset is then split into 3 sets for training this model, as training dataset, validation dataset and a test dataset. As their names suggests, the training dataset is used for training the model, the validation dataset is used for cross-validation

of the trained model during the training process after every epoch and the test dataset is used to predict if the trained model is working as expected or not. The model uses the adam optimiser and the sparse_categorical_crossentropy loss function and the accuracy as metrics for training the model. Upon successfully training the dataset along with cross-validating it after every epoch, the model is trained after 5 epochs as there is not significant improvement in the performance of the model on increasing the number of epochs thereafter. Thus, while testing this trained Feed Forward Neural Network (FFNN) model on the test dataset previously created, the model provides probabilities for the likelihood of all types of attacks in the classification under consideration. This is then used and further processed to calculate the type of attack with maximum probability as the prediction of the FFNN model. Upon testing the model on the test dataset, it is observed that this FFNN model gives an accuracy of 96.64% with an F1 score of 0.97 for the test dataset. It took 2404 seconds to train this FFNN model.

On the other hand, training a Logistic Rgression model using the same training data took only 24 seconds, whereas the test accuracy for it resulted in 93.13% with an F1 score of approximately 0.94. Similarly, the Decision Tree Classifier took 51 seconds to train the dataset providing almost 96.06% test accuracy with an F1 score of almost 0.96.

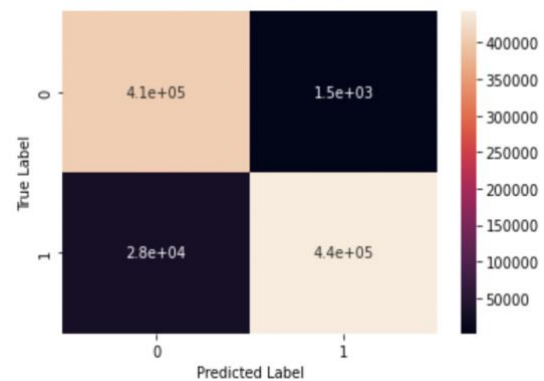


Figure 9. Confusion Matrix for Feed Forward Neural Network

Model	Accuracy	F1 score
FFNN	96.64%	0.97
Logistic Regression	93.13%	0.94
Decision Tree	96.06%	0.96
SVM	96.06%	0.96

However, through the usage of Support Vector Machine (SVM), the Support Vector Classifier (SVC) model was trained with a linear kernel, accessible through the SVM module of the scikit-learn python library, and training it using the same dataset, this model provides an accuracy of almost 96.06%. It took almost 2-3 hours to train this model on the

¹⁴ Available from

<https://www.analyticsvidhya.com/blog/2020/01/fundamental>

[s-deep-learning-activation-functions-when-to-use-them/](#)
[online]

same device while it gave an F1 score of 0.9579. Even though it took so much time to train this model and was among the best Machine Learning models with only the Decision tree Classifier close to it, the SVC model was unable to beat FFNN in terms of the test accuracy. This is a spectacular result for the FFNN model as it beat all of the other Machine Learning models in terms of the test accuracy for the classification and also its F1 score. This can be particularly described as stated by Patricia Bassey that the main reason for the difference is the approach of a SVM used for the classification process; it's main method is through finding the best margin (distance between the line and the support vectors) which differentiates the various categories of classification which further reduces the risk of error¹⁵. Moreover, it is noteworthy to point out that SVM Classifiers examines data being close to the other categories¹⁶ and also, one of its characteristics is that it works well with high dimensional space. Nevertheless, one can say that the deep learning model, Feed Forward Neural Network worked the best among all the other models through extensive understanding of the dataset through the hidden layers and then providing a better and more accurate output than all other models investigated in this research paper.

V. CONCLUSION AND FUTURE WORK

After careful consideration of the accuracy of various models and the ROC curve for the respective models, it can be said that the performance of SVM classifier is better than the Feed Forward Neural Network Model, however, the FFNN model is not too shy from similar performance. Thus, considering our dataset dimensions, it can be said that for the CSE-CIC-IDS 2018 dataset, SVM classifier did its charm and worked the best. Furthermore, this study is just with respect to the artificially curated dataset in real-time, but the study itself is static, however, the model's actual efficiency could only be measured if it is used for a software and is deployed in real-time. A more practical study must be conducted, and it should further be tested in a controlled environment to learn more about its behaviour, so that its respective pros and cons would be duly noted and worked upon accordingly to invent the best possible state-of-the-art intrusion detection system.

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¹⁵ Available from <https://medium.com/axum-labs/logistic-regression-vs-support-vector-machines-svm-c335610a3d16> [online]

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